Data Science for Mathematicians

Case Study: Predicting Medical Insurance Charges

Department of Mathematics and Computer Science

Problem Statement

Goal: Predict individual medical insurance charges (in USD) from policyholder characteristics.

Features available:

- Age, sex, body mass index (BMI)
- Number of dependants (children)
- Smoking status
- Geographic region

Task type: Regression ($y \in \mathbb{R}$, continuous target).

Data Science Workflow

We follow a systematic nine-step pipeline:

1. Data loading and initial inspection
2. Data cleaning (duplicates, missing values)
3. Exploratory data analysis (EDA)
4. Feature engineering and data splitting
5. Model training with hyperparameter tuning
6. Model evaluation on held-out test set
7. Model comparison and selection
8. Model deployment (serialisation)
9. Conclusion and future directions

Dataset Overview

The dataset `medical_insurance.csv` contains 2,772 records with 7 columns:

Feature	Type	Description
`age`	Numerical	Policyholder age (18–64)
`sex`	Categorical	Female / Male
`bmi`	Numerical	Body mass index
`children`	Numerical	Number of dependants (0–5)
`smoker`	Categorical	Yes / No
`region`	Categorical	NE, NW, SE, SW
`charges`	Numerical	Target variable (USD)

Descriptive Statistics

	age	bmi	children	charges
count	2,772	2,772	2,772	2,772
mean	39.11	30.70	1.10	13,261.37
std	14.08	6.13	1.21	12,151.77
min	18.00	15.96	0	1,121.87
50%	39.00	30.45	1	9,333.01
max	64.00	53.13	5	63,770.43

- Mean BMI $\approx$ 30.7 ("obese" category by WHO classification).
- Charges have large standard deviation ($12,152) relative to the mean ($13,261), indicating substantial variability.

Data Cleaning

Missing values: None found in any column.

Duplicate removal:

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- Likely caused by concatenation of overlapping data sources.

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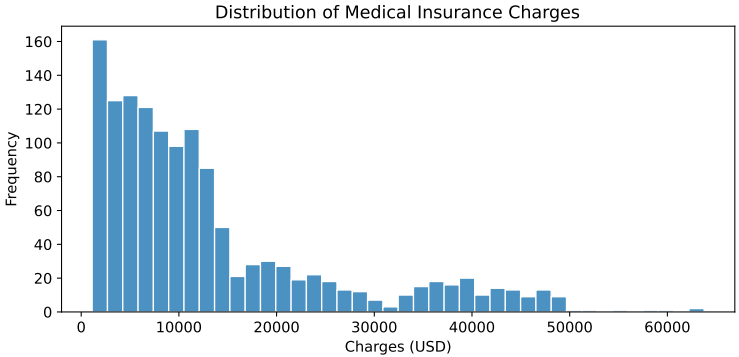
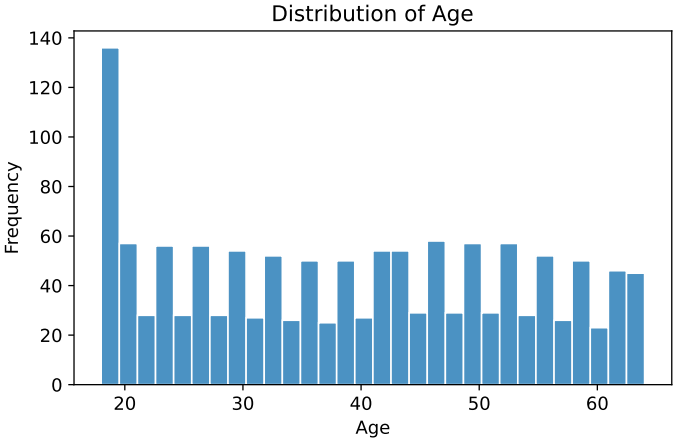
Stage	Rows
Before cleaning	2,772
Duplicates removed	1,435
After cleaning	**1,337**

Target Distribution



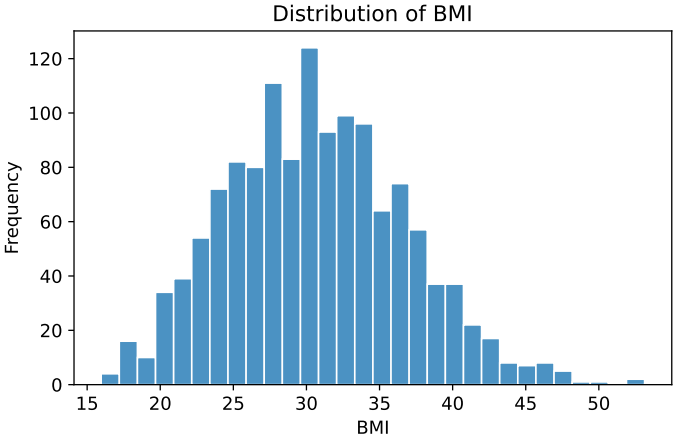
- Right-skewed distribution (skewness = 1.52).
- Mean: \$13,279 Median: \$9,386 Max: \$63,770.

Numerical Feature: Age



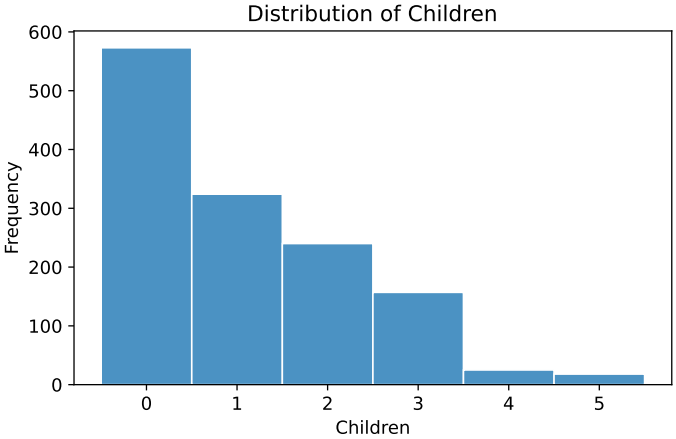
Age spans 18–64 years with a relatively uniform spread, ensuring balanced representation across age groups.

Numerical Feature: BMI



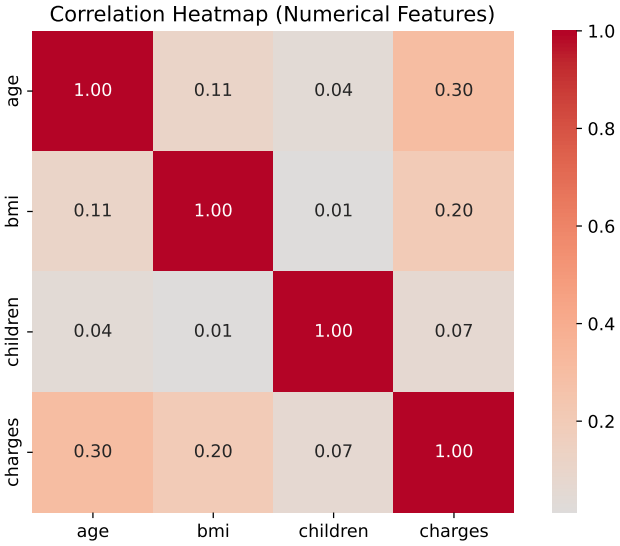
BMI follows a roughly normal distribution with mean 30.70 and standard deviation 6.13. The centre lies in the "obese" range (BMI $\geq$ 30).

Numerical Feature: Children

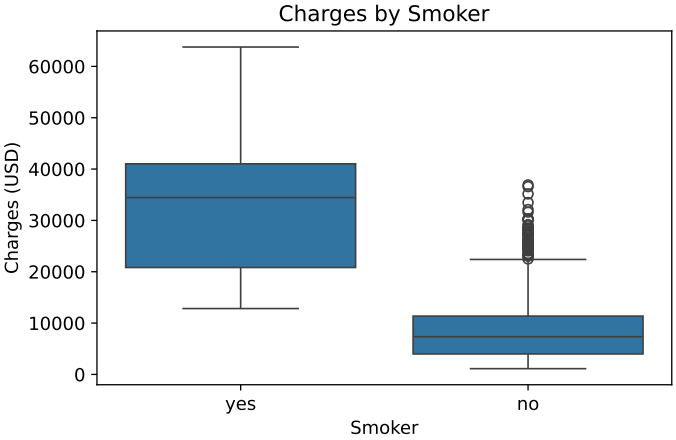


Discrete variable ranging from 0 to 5. Most policyholders have zero or one dependant, with frequency decreasing monotonically.

Correlation Heatmap

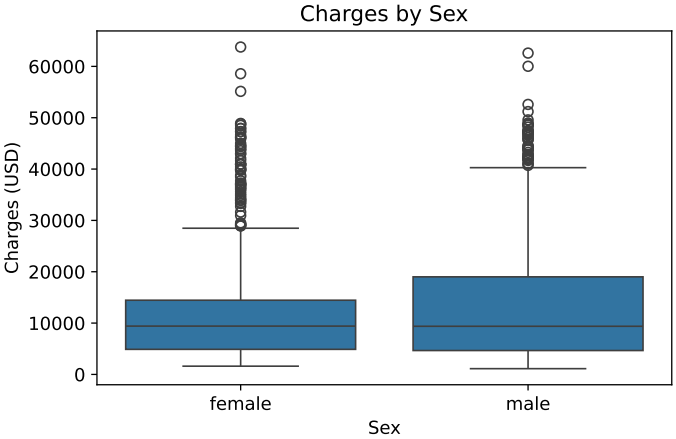


Charges by Smoking Status



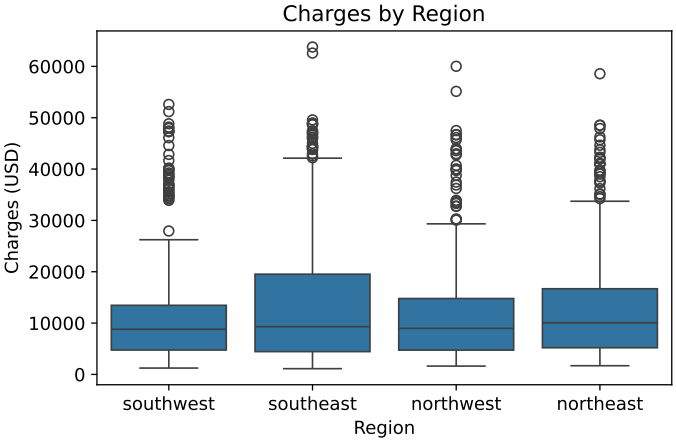
Smoking status produces the most striking separation: median charge for smokers is approximately **three times** higher than for non-smokers.

Charges by Sex



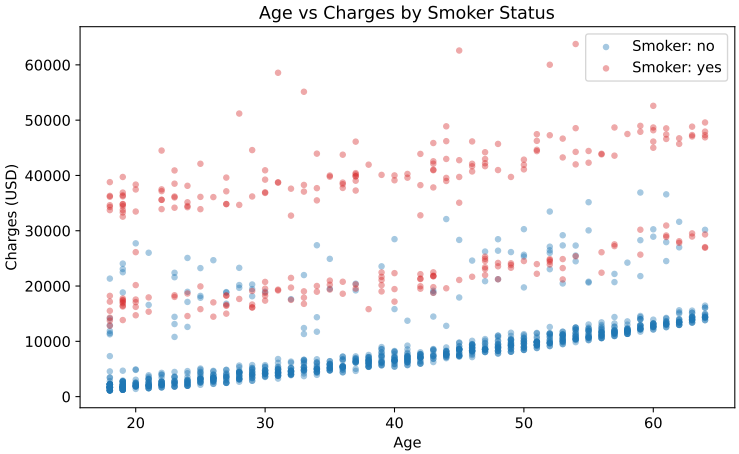
Distributions for female and male policyholders are nearly identical. Sex alone is a weak predictor.

Charges by Region



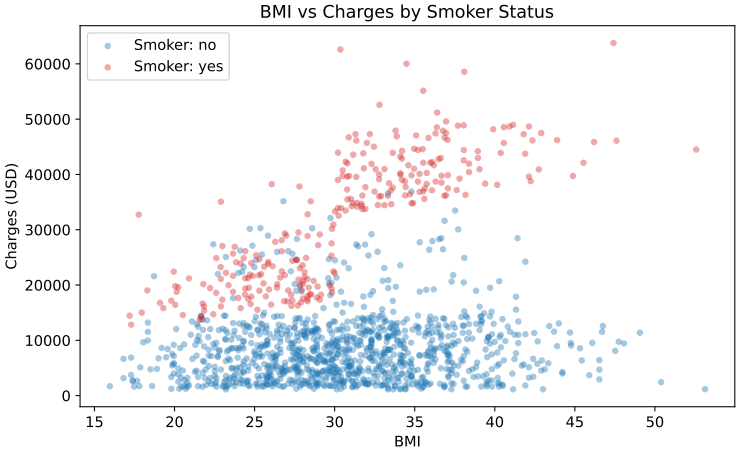
All four regions exhibit similar median charges and spread. Region is unlikely to be a dominant predictor.

Age vs Charges by Smoker Status



Two approximately linear bands emerge: non-smokers (lower, gentler slope) and smokers (higher, steeper slope).

BMI vs Charges by Smoker Status



Among non-smokers, BMI shows little relationship with charges. Among smokers, higher BMI drives charges up steeply. This motivates a **smoker–BMI interaction** feature.

EDA Summary

Key findings:

1. Target is right-skewed (skewness $= 1.52$).
2. **Smoking status** is the dominant predictor of charges.
3. BMI effect is conditional on smoking: strong for smokers, negligible for non-smokers.
4. Age correlates linearly with charges in both groups.
5. Sex and region have minimal predictive power individually.

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Action: Engineer a smoker–BMI interaction feature to capture the nonlinear relationship.

Interaction Term: Smoker $\times$ BMI

Definition: Smoker–BMI interaction

For observation i with smoking indicator $s_i \in \{0, 1\}$ and body mass index b_i:

$$x_{\text{smoker_bmi},i} = s_i \cdot b_i.$$

- For non-smokers: $x_{\text{smoker_bmi}} = 0$.
- For smokers: $x_{\text{smoker_bmi}} = \text{BMI value}$.

This captures a nonlinear relationship within a linear framework.

One-Hot Encoding and Feature Matrix

After one-hot encoding (with `drop_first=True`), the feature matrix $\mathbf{X} \in \mathbb{R}^{1337 \times 9}$:

Numerical:

1. `age`
2. `bmi`
3. `children`
4. `smoker_bmi`

Binary:

5. `sex_male`
6. `smoker_yes`
7. `region_northwest`
8. `region_southeast`
9. `region_southwest`

Target vector $\mathbf{y} \in \mathbb{R}^{1337}$ contains charge values in USD.

Train/Test Split and Feature Scaling

Data splitting:

- 80/20 ratio with fixed random state for reproducibility.
- Training set: 1,069 samples Test set: 268 samples.

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Feature scaling:

- `StandardScaler`: zero mean, unit variance.
- Fit on training set only, then transform both sets.
- Prevents information leakage from test set into preprocessing.

$$\tilde{x}_j = \frac{x_j - \bar{x}_j^{(\text{train})}}{s_j^{(\text{train})}}$$

Training Strategy

Approach: Train seven regression models, each with hyperparameter tuning via `GridSearchCV`.

- 5-fold cross-validation
- Scoring metric: R^2 (coefficient of determination)
- All models receive scaled features

Models:

1. Linear regression (baseline)
2. Ridge regression (ℓ_2 regularisation)
3. Lasso regression (ℓ_1 regularisation)
4. Support vector regression (SVR)
5. Decision tree
6. Random forest
7. Gradient boosting

Linear Regression (Baseline)

Definition: Ordinary least squares

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2$$

- No tunable hyperparameters.
- Best CV R^2: 0.8218.

Ridge Regression (ℓ_2 Regularisation)

Definition: Ridge objective

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right\}$$

- Grid: $\alpha \in \{0.1, 1.0, 10.0, 100.0\}$.
- Best: $\alpha = 0.1$, CV $R^2 = 0.8218$.
- Performance matches OLS: features are well-conditioned.

Lasso Regression (ℓ_1 Regularisation)

Definition: Lasso objective

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2 + \alpha \|\boldsymbol{\beta}\|_1 \right\}$$

- Grid: $\alpha \in \{0.1, 1.0, 10.0, 100.0\}$.
- Best: $\alpha = 1.0$, CV $R^2 = 0.8218$.
- ℓ_1 can drive coefficients to zero (feature selection), but no improvement here.

Support Vector Regression

Definition: ε-insensitive SVR

In a reproducing kernel Hilbert space $\mathcal{H}_K$:

$$\min_{f \in \mathcal{H}_K} \frac{1}{2}\|f\|^2_{\mathcal{H}_K} + C \sum_{i=1}^{n} \max\big(0, |y_i - f(\mathbf{x}_i)| - \varepsilon\big)$$

- Grid: $C \in \{10^3, 5 \times 10^3, 10^4\}$, $\varepsilon \in \{0.1, 1.0, 10.0\}$, kernel $\in$ {linear, poly, rbf, sigmoid}.
- Best: $C = 10{,}000$, $\varepsilon = 10.0$, RBF kernel.
- CV $R^2 = 0.8127$.

Decision Tree

At each node the tree selects feature j and threshold t minimising:

$$\min_{j,t} \left[\frac{n_L}{n} \mathrm{MSE}(R_L) + \frac{n_R}{n} \mathrm{MSE}(R_R) \right], \quad \mathrm{MSE}(R) = \frac{1}{|R|} \sum_{i \in R} (y_i - \bar{y}_R)^2.$$

- Grid: $d_{\max} \in \{3, 5, 7, 10\}$, $n_{\min} \in \{2, 5, 10\}$.
- Best: $d_{\max} = 5$, $n_{\min} = 2$.
- CV $R^2 = 0.8282$.

Random Forest

Definition: Random forest prediction

Ensemble of B decorrelated decision trees, each fitted on a bootstrap sample:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}).$$

- Grid: $B \in \{100, 200\}$, $d_{\text{max}} \in \{5, 10\}$, $n_{\text{leaf}} \in \{1, 4\}$.
- Best: $B = 200$, $d_{\text{max}} = 5$, $n_{\text{leaf}} = 4$.
- **Highest CV** $R^2 = 0.8444$.

Gradient Boosting

Definition: Gradient boosting update

Sequential construction of M trees correcting residual errors:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta\, h_m(\mathbf{x}), \quad m = 1, \ldots, M,$$

where h_m is fitted to the negative gradient of the loss and η is the learning rate.

- Grid: $M \in \{100, 200\}$, $\eta \in \{0.05, 0.1\}$, $d_{\max} \in \{3, 5\}$.
- Best: $M = 100$, $\eta = 0.05$, $d_{\max} = 3$.
- CV $R^2 = 0.8425$.

Cross-Validation Results Summary

Model	CV R^2
Random Forest	**0.8444**
Gradient Boosting	0.8425
Decision Tree	0.8282
Ridge Regression	0.8218
Linear Regression	0.8218
Lasso Regression	0.8218
SVR	0.8127

Ensemble methods (random forest, gradient boosting) outperform single models in cross-validation.

Evaluation Metrics

We evaluate each tuned model on the held-out test set using two metrics:

Definition: Mean absolute error

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Definition: Coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Test Set Performance

Model	Test MAE (USD)	Test R^2
Random Forest	2,413.46	**0.9012**
Gradient Boosting	2,463.99	0.9011
Decision Tree	2,681.69	0.8915
Ridge Regression	2,829.04	0.8863
Linear Regression	2,828.97	0.8862
Lasso Regression	2,828.98	0.8862
SVR	**1,875.34**	0.8790

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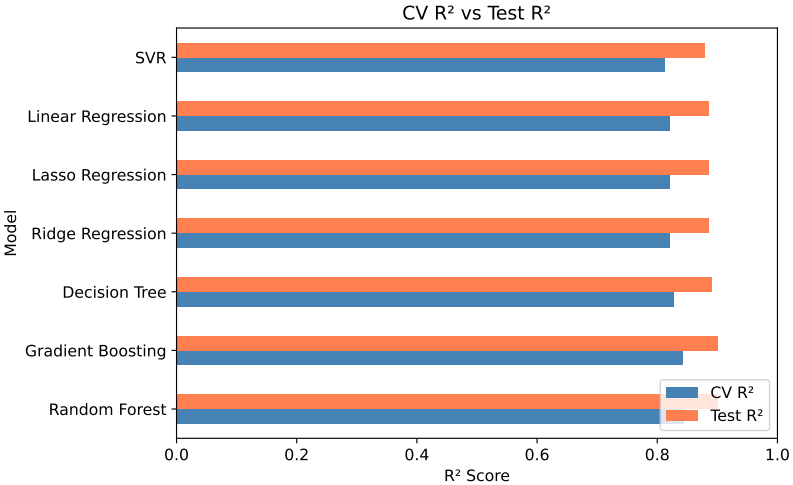
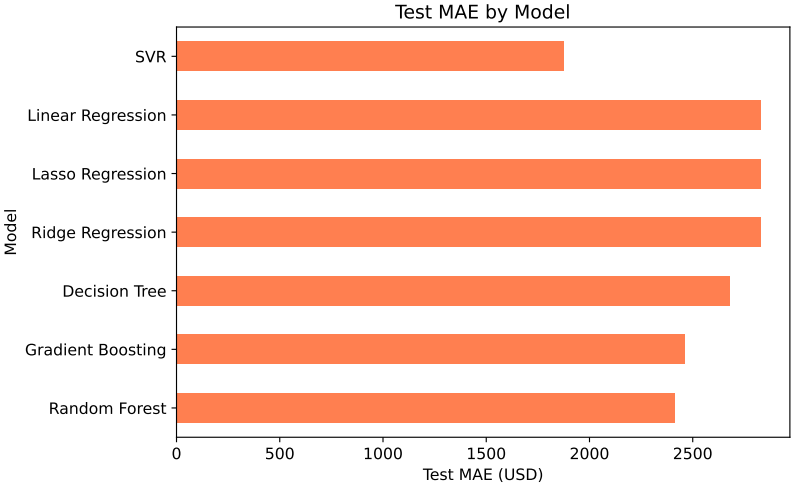
- Random forest achieves the highest R^2 (0.9012).
- SVR achieves the lowest MAE ($1,875) but the lowest R^2.
- Different metrics can lead to different model rankings.

CV R^2 vs Test R^2



Every model achieves higher R^2 on the test set than in CV—the particular test split is slightly

Test MAE by Model



SVR's low MAE reflects its ε-insensitive loss, which is robust to outliers. R^2 penalises large

Full Model Comparison

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Selected model: Random forest — highest test R^2 of 0.9012, explaining over 90% of the variance in unseen charges.

MAE vs R^2: A Trade-off

Why do MAE and R^2 rank models differently?

- R^2 is based on **squared** residuals: large errors are penalised disproportionately.
- MAE uses **absolute** residuals: all errors contribute equally.

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- SVR's ε-insensitive tube ignores small errors, focusing on reducing typical deviations.
- Random forest minimises MSE-based splits, achieving better overall variance explanation.

Lesson: Always evaluate multiple metrics; the "best" model depends on the application context.

Model Serialisation

We bundle the trained pipeline into a single artifact using `pickle`:

Key	Content
`model`	Fitted `RandomForestRegressor`
`model_name`	`"Random Forest"`
`scaler`	Fitted `StandardScaler`
`features`	Ordered list of 9 feature names

Bundling model, scaler, and feature list ensures reproducible preprocessing and prediction without access to the training data.

Inference on a New Observation

Example: Prediction for a new policyholder

A 45-year-old male smoker, BMI $= 28.5$, 2 children, southeast region.

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1. Compute interaction: `smoker_bmi` $= 1 \times 28.5 = 28.5$.
2. One-hot encode categorical features.
3. Scale using stored scaler statistics.
4. Predict: $\hat{y} = \$26{,}325.57$.

The high predicted cost is consistent with the strong influence of smoking and the smoker–BMI interaction.

Summary of Results

Data cleaning:

- 2,772 rows $\rightarrow$ 1,337 unique records after removing 1,435 duplicates.

EDA findings:

- Right-skewed target (skewness $= 1.52$).
- Smoking status is the dominant predictor; BMI effect is conditional on smoking.

Best model:

- Random forest: test $R^2 = 0.9012$, test MAE $=$ \$2,413.46.
- Explains over 90% of the variance in unseen charges.

Practical Implications

- The strong influence of smoking on predicted charges aligns with actuarial knowledge.
- The smoker–BMI interaction suggests that **targeted health interventions** for smokers with high BMI could yield the largest reduction in insurance costs.
- The exported pickle artifact enables **real-time charge estimation** for new policyholders.

Limitations

- **Small dataset:** Only $n = 1{,}337$ unique records after deduplication, limiting model complexity.
- **Data quality:** The large number of duplicates raises questions about data collection methodology.
- **Missing predictors:** Important features like pre-existing conditions, medication usage, and lifestyle factors are absent.

Future Work

1. Collect a larger, deduplicated dataset with richer features (pre-existing conditions, prescriptions).
2. Explore nonlinear target transformations (e.g., log-transform) to address skewness.
3. Test additional ensemble methods (XGBoost, LightGBM) with more extensive hyperparameter grids.
4. Incorporate domain knowledge through feature engineering (age-group binning, WHO BMI categories).

Key Takeaways

1. **Data cleaning matters:** Over half the rows were duplicates—always inspect before modelling.
2. **EDA drives feature engineering:** The smoker–BMI interaction was discovered through visual exploration and significantly improved model performance.
3. **Compare multiple models:** Seven models were trained and tuned; the best (random forest, $R^2 = 0.90$) was not obvious *a priori*.
4. **Metrics tell different stories:** SVR had the best MAE but worst R^2—always evaluate multiple metrics.
5. **Deploy reproducibly:** Bundle model, scaler, and feature list into a single serialised artifact.